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On strong odd colorings of graphs

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ABSTRACT

A strong odd coloring of a simple graph G is a proper coloring of the vertices of G such that for every vertex v and every color c, either c is used an odd number of times in the open neighborhood $N_G(v)$ or no neighbor of v is colored by c. The smallest integer k for which G admits a strong odd coloring with k colors is the strong odd chromatic number, $\chi_{so}(G)$. These coloring notion and graph parameter were recently defined in Kwon and Park (arXiv:2401.11653). We answer a question raised by the originators concerning the existence of a constant bound for the strong odd chromatic number of all planar graphs. We also consider strong odd colorings of trees, unicyclic graphs, claw-free graphs, and graph products.

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1. Introduction

All graphs considered here are finite, loopless and undirected. If there are no parallel edges we speak of a *simple* graph. Let $G = (V_G, E_G)$ be a simple graph with vertex set V_G and edge set E_G. For a vertex v, we use $N_G(v)$ to denote the open neighborhood of v, and $\deg_G(v)$ to denote the degree of v, i.e., the number of edges of G incident with v. A *k-vertex*, a *k^--vertex*, and a *k^+-vertex*, respectively, is a vertex of degree k, of degree at most k, and of degree at least k. Let $\Delta(G)$ denote the maximum degree of G. The maximum average degree $\mathrm{mad}(G)$ of G is the maximum of $\frac{2|E_H|}{|V_H|}$ over all non-void subgraphs H of G. The girth, $g(G)$, is the length of a shortest cycle in G. A graph (as a combinatorial object) is said to be *planar* if it can be embedded in the Euclidean plane (or equivalently on the sphere) without edges crossing; such a drawing is called a *plane* graph (a geometrical object).

A k-(vertex-)coloring of a graph G is an assignment $\varphi : V_G \to \{1, \ldots, k\}$. A coloring φ is said to be *proper* if every color class (i.e., preimage $\varphi^{-1}(c)$ of any color c) is an independent subset of the vertex set of G. The minimum k for which a graph G admits a proper k-coloring is the *chromatic number* of G, denoted $\chi(G)$. A *square coloring* of a graph G is a proper coloring of G^2, the latter being the graph obtained from G by linking with an edge each pair of non-adjacent vertices that have a common neighbor (in other words, vertices that are at distance 2 in G). In 1977, Wegner conjectured an upper bound for $\chi(G^2)$ when G is planar.

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Conjecture 1.1 *(Wegner [31]). Let G be a planar graph. Then*

$$\chi(G^2) \le \begin{cases} 7 & \text{if } \Delta(G) = 3\,, \\ \Delta(G) + 5 & \text{if } 4 \le \Delta(G) \le 7\,, \\ \left\lfloor \frac{3\Delta(G)}{2} \right\rfloor + 1 & \text{if } \Delta(G) \ge 8. \end{cases}$$

Thomassen [28] and Hartke et al. [18] independently proved Conjecture 1.1 when $\Delta(G) = 3$. For related research on square coloring problems and best known bounds on $\chi(G^2)$ see [4,5,13,14].

A recently introduced intermediate coloring concept, which can be seen as both a weakening of square coloring and a strengthening of proper coloring, has attracted considerable attention among researchers. Defined by Petruševski and Škrekovski [25], an *odd coloring* of G is a proper coloring such that for any non-isolated vertex v in G, there is a color that is used an odd number of times on the neighborhood of v. The *odd chromatic number* of G, denoted $\chi_o(G)$, is the minimum number of colors in an odd coloring of G. Determining the odd chromatic number of a graph and deciding whether a bipartite graph is odd k-colorable for $k \ge 3$ are two $\mathcal{NP}$-complete problems [1]. Petruševski and Škrekovski [25] showed that all planar graphs are odd 9-colorable, and conjectured the following.

Conjecture 1.2 *(Petruševski and Škrekovski [25]). For every planar graph G, it holds that $\chi_o(G) \le 5$.*

Note that a 5-cycle is a planar (moreover, an outerplanar) graph whose odd chromatic number is exactly 5. A study on various aspects of the odd chromatic number was carried out by Caro, Petruševski and Škrekovski [6]. Building upon some results in [6], Petr and Portier [24] proved that planar graphs are odd 8-colorable. Besides planar graphs with bounded girth, some special embedded graphs have also been investigated, such as 1-planar graphs [9,22], outerplanar graphs [6], outer-1-planar graphs and 2-boundary planar graphs [26], and k-planar graphs [19,15]. Tian and Yin [29] showed that every toroidal graph with girth at least 4 is odd 7-colorable. Odd colorings of sparse graphs, measured in terms of the maximum average degree, were investigated by Cranston [8] and Cho, Choi, Kwon, and Park [7]. Their findings include that every planar graph of girth at least 5, 7 or 11, respectively, is odd 6-, 5- or 4-colorable. Most recently, Miao, Sun, Tu and Yu [23] gave support to Conjecture 1.2 by showing that all triangle-free planar graphs without intersecting 4-cycles are odd 5-colorable. They also showed that planar graphs with non-intersecting (resp. edge-disjoint) 3-cycles are odd 6-colorable (resp. odd 7-colorable).

In this paper, we delve into a natural stronger variant of the odd coloring notion, introduced of late by imposing a stronger neighborhood constraint on the properness condition. Defined by Kwon and Park [21], a *strong odd coloring* of a graph G is a proper coloring of its vertices such that for every non-isolated vertex v, if a color c is used on $N_G(v)$ then c appears an odd number of times on the vertices comprising $N_G(v)$. Clearly, the proper setting for any study of this coloring concept deals with simple graphs, since parallel edges are irrelevant for either requirement. Every simple graph admits a strong odd coloring since a square coloring is also a strong odd coloring. The *strong odd chromatic number* of G, denoted $\chi_{so}(G)$, is the minimum k such that G admits a strong odd k-coloring. It is clear from the definitions that for every graph G,

$$\chi(G) \le \chi_o(G) \le \chi_{so}(G) \le \chi(G^2) \le (\Delta(G))^2 + 1\,.$$

However, as observed in [21], the difference $\chi(G^2) - \chi_{so}(G)$ can be arbitrarily large: indeed, if $G = K_{m,n}$ then $\chi(G^2) = m + n$ whereas $\chi_{so}(G) \le 4$. The originators of the concept investigated strong odd colorings of sparse graphs G. They proved that if G has $\mathrm{mad}(G) \le \frac{20}{7}$ then $\chi_{so}(G) \le \Delta(G) + 4$; moreover, the bound is tight for subcubic planar graphs. Also if G is a graph with $\mathrm{mad}(G) \le \frac{30}{11}$ and $\Delta(G) \ge 4$ then $\chi_{so}(G) \le \Delta(G) + 3$. Since results regarding bounded maximum average degree have natural corollaries to planar graphs with girth restrictions, because the class of graphs G with $\mathrm{mad}(G) < \frac{2g}{g-2}$ includes all planar graphs with girth at least g, the findings of this pair of scholars yield that every planar graph G with girth $g \ge 7$ (resp. $g \ge 8$) has $\chi_{so}(G) \le \Delta(G) + 4$ (resp. $\chi_{so}(G) \le \Delta(G) + 3$). In view of the odd 8-colorability of all planar graphs [24], Kwon and Park ended [21] with the following natural question.

Question 1.3. *Is there a constant C such that every planar graph G has $\chi_{so}(G) \le C$?*

One of the main contributions of the present work is a proof that such a finite upper bound C exists.

This paper is organized as follows. In Section 2, we provide bounds for the strong odd chromatic number of trees, unicyclic graphs, and claw-free graphs. In Section 3, we consider planar and outerplanar graphs in the same regard and answer Question 1.3 in the affirmative. In Section 4, we study strong odd coloring of graph products.

2. Trees, unicyclic graphs, and claw-free graphs

Let us call a tree *odd* if the degree of every vertex is odd.

Proposition 2.1. *For every tree T we have $\chi_{so}(T) \leq 3$, and $\chi_{so}(T) = 2$ holds if and only if T is odd. Moreover, a strong odd coloring of T with $\chi_{so}(T)$ colors can be determined in linear time.*

Proof. If T is odd, then its unique proper 2-coloring is a strong odd coloring, hence $\chi_{so}(T) = 2$ and of course T can be properly 2-colored in linear time. Otherwise two colors are not enough, unless T is trivial, because if a vertex v has a positive even degree $\deg_T(v)$, then the closed neighborhood $N_T[v] = N_T(v) \cup \{v\}$ requires at least three colors under any strong odd coloring of T. We give the following linear-time algorithm to show that three colors are always enough.

Choose a root vertex r and traverse T with Breadth-First Search. Let the root r have color 1. If $\deg_T(r)$ is odd, assign color 2 to all of $N_T(r)$; contrarily, color one neighbor of r with 3 and the rest of $N_T(r)$ with 2. Afterwards, at each non-root vertex v—in the order as they are visited during Breadth-First Search—if the number of its children is even, make the entire $N_T(v)$ monochromatic. Contrarily, if v has an odd number of children, then let one child of v be assigned with the unique color from $\{1, 2, 3\}$ which is distinct from the colors of v and of its parent, and let all the other children of v receive the color of the parent of v. At the end of this procedure the entire T is strongly odd 3-colored. □

Proposition 2.2. *If G is a connected unicyclic graph other than C_5, then $\chi_{so}(G) \leq 4$. Moreover, a strong odd 4-coloring of G can be determined in linear time.*

Proof. It is easy to check that for cycles we have:

$$\chi_{so}(C_n) = \begin{cases} 3 & \text{if } 3 \mid n\,; \\ 4 & \text{if } 3 \nmid n \text{ and } n \neq 5\,; \\ 5 & \text{if } n = 5\,. \end{cases} \tag{1}$$

Moreover, a strong odd $\chi_{so}(C_n)$-coloring of C_n can be determined in linear time.

Assume now that G is a connected unicyclic graph which is not a cycle. Recall that, by Proposition 2.1, every tree can be strongly odd 3-colored in linear time. Suppose first that the cycle in G is some C_n with $n \neq 5$. Then, by (1), there exists a strong odd 4-coloring φ of the vertices of this cycle. Consider a vertex $a \in C_n$ that has neighbors outside the cycle, and let b and c be the neighbors of a on the cycle. Assume without loss of generality that $\varphi(a) = 1$, $\varphi(b) = 2$ and $\varphi(c) = 3$ (these must be three distinct colors due to the strong coloring of the cycle). Let $N^*(a) = N_G(a) \setminus \{b, c\}$ be the set of all neighbors of a not on the cycle. If $N^*(a)$ is even-sized, then color it monochromatically with 2. Contrarily, if $N^*(a)$ is odd-sized, color it monochromatically with 4. Note that this partial coloring is still strong odd. Now each vertex in $N^*(a)$ is the root of a tree that emanates from it away from the cycle. Each such tree can be colored by three colors avoiding the color 1 (which is used for a); so one can use the colors $2, 3, 4$ and thus complete a strong odd 4-coloring of G. Therefore, unless the cycle of G is C_5, four colors suffice for a strong odd coloring of G and such a coloring can be determined in linear time.

Suppose now that the cycle in G is C_5 with vertices a, b, c, d, e in a cyclic order so that the vertex a has neighbors outside the cycle. Let $N^*(a) = N_G(a) \setminus \{b, e\}$. Assign colors as follows: $\varphi(a) = 1$, $\varphi(b) = 2$, $\varphi(c) = 3$, $\varphi(d) = 4$, $\varphi(e) = 5$, the last one considered as a temporary color only. Again, we distinguish between two possibilities regarding the parity of $|N^*(a)|$.

Case 1: *$N^*(a)$ is even-sized.* Split $N^*(a)$ into two odd-sized subsets X and Y (say $|X| = |N^*(a)| - 1$, $|Y| = 1$). Recolor the vertex e by 2. Color all of X by 2 and all of Y by 3. This is still a (partial) strong odd coloring but with four colors. Now apply to the trees emanating from $N^*(a)$ and from any other vertex of C_5 the above argument.

Case 2: *$N^*(a)$ is odd-sized.* Color all of $N^*(a)$ with 2. Again, recolor the vertex e by 2 and then complete a strong odd 4-coloring as above.

This concludes the proof that apart from C_5 every other connected unicyclic graph admits a strong odd 4-coloring which can be determined in linear time. □

The authors of [21] mentioned the following fact without proof. Below we include a short argument.

Proposition 2.3. *If G is a claw-free graph, then $\chi_{so}(G) = \chi(G^2)$.*

Proof. We know that $\chi_{so}(G) \leq \chi(G^2)$ holds for every G. Assume that G is claw-free, and suppose for a contradiction that $\chi_{so}(G) < \chi(G^2)$. Consider a strong odd coloring of G with $\chi_{so}(G)$ colors. Then there exists a vertex, say v, whose neighborhood contains some color more than once. It requires at least three occurrences of that color for the coloring to be odd. Say, the vertices x, y, z have the same color. In this case, however, $\{x, y, z\}$ must be an independent set, implying the contradiction that G contains a claw with center v and leaves x, y, z. □

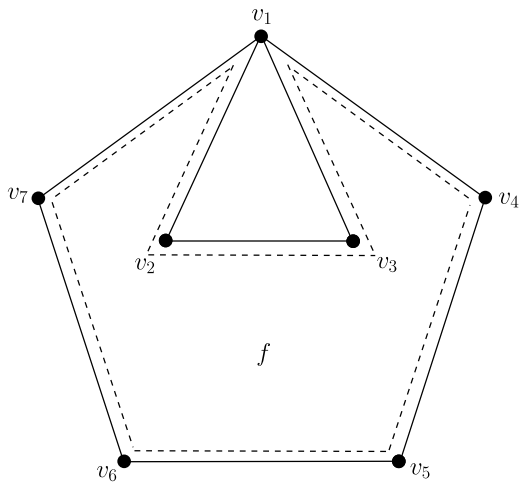

Fig. 1. The boundary walk of the face f.

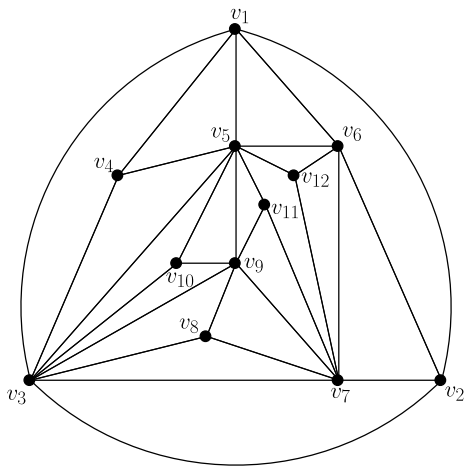

Fig. 2. The graph $G^{12.a}$.

3. Planar graphs and outerplanar graphs

Recall that a planar graph is a graph which can be embedded in the (Euclidean) plane. A plane graph is a fixed embedding of a planar graph. If a plane graph G is drawn in the plane $\mathcal{P}$, then the maximal connected regions of $\mathcal{P}\backslash G$ are the *faces* of G. Faces of G are open 2-cells. The *boundary* of a face f is the boundary in the usual topological sense, denoted $\partial(f)$. It is a collection of all vertices and edges, respectively called *boundary vertices* and *boundary edges* of f, contained in the closure of f that can be organized into a closed walk in G traversing along a simple closed curve lying just inside the face f. This closed walk is unique up to the choice of initial vertex and direction, and is called the *boundary walk* of the face f (see [17], p. 101). The sets of boundary vertices and boundary edges of f are denoted by $V(f)$ and $E(f)$, respectively. Similarly, $F(v)$ denotes the set of faces incident with a vertex v. For all notation defined, the relevant graph may be referred to as a subscript if necessary. When e_1 and e_2 are a pair of parallel edges which along with their shared endpoints constitute the boundary of a face f, such an f is called a *digon*. Whenever we are dealing with simple graphs, the boundary walk can be seen as a list of consecutive vertices along the boundary of f. As an example see the graph in Fig. 1 where $v_1, v_2, v_3, v_1, v_4, v_5.v_6, v_7, v_1$ is a boundary walk of a face f. A graph is *outerplanar* if it can be embedded in the plane such that all vertices are incident with a single face; e.g., the graph in Fig. 1 is outerplanar. (A more traditional way of drawing would place the triangle outside of the pentagon, and an equivalent standard definition of outerplanar graphs requires the existence of a planar embedding where all vertices are on the boundary of the infinite region.)

In this section we give estimates on the largest possible value of χ_{so} in the classes of planar and outerplanar graphs. Figs. 2 and 3 exhibit two planar graphs that we denote by $G^{12.a}$ and $G^{12.b}$, respectively. Each of those two graphs has order 12, diameter 2 and provides the lower bound 12 for $\max\{\chi_{so}(G) \mid G \text{ is planar}\}$. Later on in this section we shall show that

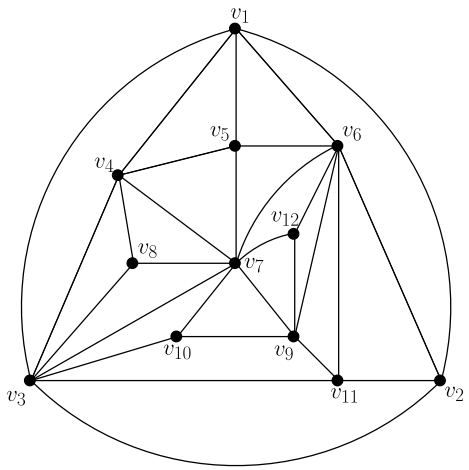

Fig. 3. The graph $G^{12.b}$.

$\max\{\chi_{so}(G) \mid G \text{ is planar}\}$ is indeed finite, but first let us discuss $G^{12.a}$ and $G^{12.b}$ in regard to their strong odd chromatic number.

Proposition 3.1. *We have* $\chi_{so}(G^{12.a}) = \chi_{so}(G^{12.b}) = 12$.

Proof. In both cases the vertices are indexed from 1 to 12. The color assignment $\varphi(v_i) = i$ obviously is a strong odd coloring, hence it will suffice to show that no two vertices can be assigned with the same color under a strong odd coloring φ. We may assume that $\varphi(v_i) \le i$ for all $1 \le i \le 12$, without loss of generality. (This situation can be attained by renumbering the colors, if necessary. Namely, for all colors i simultaneously, for all vertices in the color class $\varphi^{-1}(i)$, replace their color with $\min\{j : \varphi(v_j) = i\}$.) For the duration of this proof, by an "i-neighbor" we shall mean a neighbor assigned with the color i.

In the graph $G^{12.a}$ we proceed in the natural order $v_1, v_2, \ldots, v_{12}$ of vertices, and demonstrate that $\varphi(v_i) = i$ $(1 \le i \le 12)$.

- $\varphi(v_1) = 1$, $\varphi(v_2) = 2$, $\varphi(v_3) = 3$ — indeed, the three vertices v_1, v_2, v_3 induce a K_3 in $G^{12.a}$.
- $\varphi(v_4) = 4$ — by adjacency, $\varphi(v_4) \notin \{1, 3\}$; $\varphi(v_4) \ne 2$, otherwise v_1 would need a third 2-neighbor, but $N(v_1) = \{v_2, v_3, v_4, v_5, v_6\}$ and v_5 is adjacent to v_4 whereas v_6 is adjacent to v_2.
- $\varphi(v_5) = 5$ — by adjacency, $\varphi(v_5) \notin \{1, 3, 4\}$; $\varphi(v_5) \ne 2$, otherwise v_1 would need $\varphi(v_6) = 2$ but v_6 is adjacent to v_2 and v_5.
- $\varphi(v_6) = 6$ — by adjacency, $\varphi(v_6) \notin \{1, 2, 5\}$; $\varphi(v_6) \notin \{3, 4\}$, otherwise v_1 would have just two 3-neighbors or two 4-neighbors.
- $\varphi(v_7) = 7$ — by adjacency, $\varphi(v_7) \notin \{2, 3, 6\}$; $\varphi(v_7) \notin \{1, 4, 5\}$ because $N(v_3) = \{v_1, v_2, v_4, v_5, v_7, v_8, v_9, v_{10}\}$) and v_8, v_9, v_{11}, v_{12} are adjacent to v_7: namely, if $\varphi(v_7) \in \{1, 4, 5\}$ then v_3 would need $\varphi(v_{10}) = \varphi(v_7)$ to have that color a third time in $N(v_3)$, but (v_5, v_{10}) is an edge excluding color 5 at v_{10}, and $\varphi(v_{10}) \in \{1, 4\}$ would yield only two neighbors of v_5 in that color.
- $\varphi(v_8) = 8$ — by adjacency, $\varphi(v_8) \notin \{3, 7\}$; $\varphi(v_8) \notin \{1, 2, 4\}$ as otherwise $\varphi(v_8) = \varphi(v_{10}) = \varphi(v_{11})$ would follow (first for $\varphi(v_{10})$ by vertex v_3, second for $\varphi(v_{11})$ by vertex v_9), but then the number of neighbors of v_5 and v_7 in that color would not have the same parity, hence a strong odd coloring could not be completed at vertex v_{12}; $\varphi(v_8) = 5$ would occur only twice in $N(v_3)$; $\varphi(v_8) = 6$ would imply $\varphi(v_{11}) = 6$ in $N(v_7)$ and then $\varphi(v_{10}) = 6$ in both $N(v_5)$ and $N(v_9)$, a contradiction in $N(v_3)$.
- $\varphi(v_9) = 9$ — by adjacency, $\varphi(v_9) \notin \{3, 5, 7, 8\}$; $\varphi(v_9) \notin \{1, 2, 4\}$ due to $N(v_3)$ and the presence of the edge (v_9, v_{10}); $\varphi(v_9) \ne 6$ due to $N(v_7)$ as v_{11} is adjacent to v_9 and v_{12} is adjacent to v_6.
- $\varphi(v_{10}) = 10$ — by adjacency, $\varphi(v_{10}) \notin \{3, 5, 9\}$; $\varphi(v_{10}) \notin \{1, 2, 4, 7, 8\}$ due to $N(v_3)$; $\varphi(v_{10}) \ne 6$ as otherwise the parity conflict between $N(v_5)$ and $N(v_7)$ concerning color 6 could not be resolved at vertex v_{11}.
- $\varphi(v_{11}) = 11$ — by adjacency, $\varphi(v_{11}) \notin \{5, 7, 9\}$; $\varphi(v_{11}) \notin \{1, 2, 4, 8, 10\}$ as any of those colors at vertex v_{11} would cause a parity conflict between $N(v_5)$ and $N(v_7)$; $\varphi(v_{11}) \ne 3$ due to $N(v_9)$; $\varphi(v_{11}) \ne 6$ as otherwise (since (v_6, v_{12}) is an edge) both v_5 and v_7 would have just two 6-neighbors.
- $\varphi(v_{12}) = 12$ — every vertex v_i $(1 \le i \le 11)$ not adjacent to v_{12} has a common neighbor v_j with v_{12}, hence assigning $\varphi(v_{12}) = i$ would yield that color i occurs precisely twice in $N(v_j)$.

The analysis for $G^{12.b}$ is based on similar principles. Again we proceed in the natural order $v_1, v_2, \ldots, v_{12}$ of vertices.

- $\varphi(v_1) = 1$, $\varphi(v_2) = 2$, $\varphi(v_3) = 3$ since $\{v_1, v_2, v_3\}$ induces a K_3 in $G^{12.b}$.
- $\varphi(v_4) = 4$ — by adjacency, $\varphi(v_4) \notin \{1, 3\}$; $\varphi(v_4) \neq 2$, as otherwise v_1 requires the color 2 on either v_5 or v_6, and both are impossible because of the edges (v_4, v_5) and (v_2, v_6).
- $\varphi(v_5) = 5$ — by adjacency, $\varphi(v_5) \notin \{1, 4\}$; $\varphi(v_5) \notin \{2, 3\}$ by $N(v_1)$ and the presence of the edge (v_5, v_6).
- $\varphi(v_6) = 6$ — by adjacency, $\varphi(v_6 \notin \{1, 2, 5\})$; $\varphi \notin \{3, 4\}$ by $N(v_1)$.
- $\varphi(v_7) = 7$ — by adjacency, $\varphi(v_7) \notin \{3, 4, 5, 6\}$; $\varphi(v_7) \neq 2$ by $N(v_6)$ and the presence of the edge (v_2, v_{11}); $\varphi(v_7) \neq 1$ by $N(v_5)$.
- $\varphi(v_8) = 8$ — by adjacency, $\varphi(v_8) \notin \{3, 4, 7\}$; $\varphi(v_8) \neq 6$ as otherwise $\varphi(v_{10}) = 6$ by $N(v_7)$ and the edges (v_6, v_{12}) and (v_9, v_{12}), however then v_9 would have precisely two 6-neighbors; $\varphi(v_8) \neq 5$ by $N(v_4)$; $\varphi(v_8) \neq 2$ as otherwise $\varphi(v_{10}) = 2$ by $N(v_3)$, but then in view of $N(v_7)$ the color 2 is used for either v_9 or v_{12}, which would force that the color 2 occurs exactly twice on $N(v_6)$; finally, $\varphi(v_8) \neq 1$ by $N(v_4)$.
- $\varphi(v_9) = 9$ — by adjacency, $\varphi(v_9) \notin \{6, 7\}$; $\varphi(v_9) \notin \{4, 5, 8\}$ by $N(v_7)$ and the edges (v_9, v_{10}) and (v_9, v_{12}); $\varphi(v_9) \neq 3$ by $N(v_{10})$; $\varphi(v_9) \notin \{1, 2\}$ by $N(v_6)$ and (v_9, v_{12}).
- $\varphi(v_{10}) = 10$ — by adjacency, $\varphi(v_{10}) \notin \{3, 7, 9\}$; $\varphi(v_{10}) \neq 8$ since otherwise $N(v_3)$ would give $\varphi(v_{11}) = 8$ and consequently $\varphi(v_{12}) \neq 8$ by $N(v_6)$, however that would create for v_7 exactly two 8-neighbors; $\varphi(v_{10}) \neq 6$ by $N(v_9)$; $\varphi(v_{10}) \notin \{4, 5\}$ as otherwise $N(v_7)$ would imply that $\varphi(v_{12}) = \varphi(v_{10})$, but then that color would appear exactly twice on $N(v_9)$; $\varphi(v_{12}) \notin \{1, 2\}$ as otherwise $\varphi(v_{11}) = \varphi(v_{10})$ by $N(v_3)$, which would give that precisely two neighbors of v_6 use the color $\varphi(v_{10})$.
- $\varphi(v_{11}) = 11$ — by adjacency, $\varphi(v_{11}) \notin \{2, 3, 6\}$; $\varphi(v_{11}) \notin \{1, 4, 7, 8, 10\}$ by $N(v_3)$; $\varphi(v_{11}) \notin \{5, 9\}$ as otherwise $\varphi(v_{12}) = \varphi(v_{11})$ by $N(v_6)$, but then $N(v_7)$ uses the color $\varphi(v_{11})$ exactly twice.
- $\varphi(v_{12}) = 12$ — every vertex v_i $(1 \leq i \leq 11)$ not adjacent to v_{12} has a common neighbor v_j with v_{12}, hence assigning $\varphi(v_{12}) = i$ would yield that color i occurs precisely twice in $N(v_j)$. □

On a similar note, we establish a lower bound for the parameter χ_{so} over the class of outerplanar graphs.

Proposition 3.2. *The outerplanar graph $G_7 = K_1 + P_6$ has $\chi_{so}(G_7) = 7$.*

Proof. The vertex from K_1, say v_1, must have its private color as it is completely adjacent to P_6. If a color appears more than once in P_6, then it has to occur at least three times. But then there is a vertex in P_6 whose both neighbors in P_6 have this color, and the color of its third neighbor v_1 in G_7 is distinct, a contradiction. Thus, each of the seven vertices has its private color. □

Let us now turn to answering the Question 1.3. We provide an affirmative answer by making use of a related concept, initiated in 2009 by Czap and Jendrol' [10]. An equivalent form of their definition reads as follows. A *facially odd coloring* (resp. *proper facially odd coloring*) of a 2-connected plane multigraph G is a coloring (resp. proper coloring) of its vertices such that every face is incident with zero or an odd number of vertices of each color. The minimum number of colors in a facially odd (resp. proper facially odd) coloring of G is denoted $\chi_{fo}(G)$ (resp. $\chi_{pfo}(G)$). Clearly, both parameters are well-defined for any plane multigraph; the confinement to 2-connected multigraphs is due to the original definition of these two coloring notions which instead of incidences of any color to a face counts for the occurrences of any color along each boundary walk of the face. Note that for any plane multigraph G we have $\chi_{fo}(G) \leq \chi_{pfo}(G)$. Czap, Jendrol' and Voigt [12] proved that every 2-connected plane multigraph G has $\chi_{pfo}(G) \leq 118$. The bound 118 was improved to 97 by Kaiser, Rucký, Stehlík and Škrekovski [20], which is the current best general upper bound on this graph parameter for the class of 2-connected plane multigraphs. On the other end of the spectrum, recently, Štorgel [27] pointed out to an infinite family of 2-connected plane graphs G with $\chi_{pfo}(G) = 12$; so the general upper bound for $\chi_{pfo}(G)$ is between 12 and 97. Bounds of $\chi_{pfo}(G)$ for 3-connected plane graphs having the property that all faces of a certain size are in a sense far from each other were considered in [11]. Wang, Finbow and Wang [30] proved that only two 2-connected outerplane graphs need 10 colors, whereas the rest admit a proper facially odd coloring with at most 9 colors. As for (possibly improper) facially odd colorings of plane graphs, Czap and Jendrol' [10] conjectured that $\chi_{fo}(G) \leq 6$ for any 2-connected plane graph G, providing an infinite family of examples having $\chi_{fo} = 6$. This conjecture was refuted by Kaiser et al. [20], by showing that the best possible general upper bound on $\chi_{fo}(G)$ for the class of 2-connected plane simple graphs G is at least 8. We define

$$\chi_{pfo}(\mathcal{P}) = \max_{G \hookrightarrow \mathcal{P}} \chi_{pfo}(G)$$

as the maximum of $\chi_{pfo}(G)$ over all 2-connected multigraphs G embedded into the plane $\mathcal{P}$; and, analogously,

$$\chi_{pfo}(\mathcal{P}_o) = \max_{G \text{ is outerplanar}} \chi_{pfo}(G)$$

for 2-connected outerplanar graphs. As mentioned in the previous paragraph, it has already been established that $12 \leq \chi_{pfo}(\mathcal{P}) \leq 97$; moreover, the tight result on outerplanar graphs is $\chi_{pfo}(\mathcal{P}_o) = 10$.

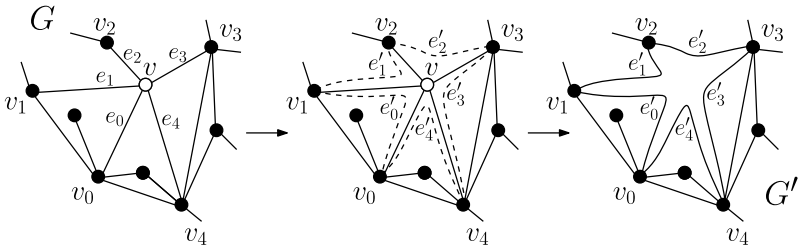

Fig. 4. The annihilation of a vertex v. The original graph G is on the left, the resulting graph G' on the right. The annihilated vertex is depicted as an empty circle.

Note that over the class of 2-connected plane multigraphs, the best general upper bound for (not necessarily proper) facially odd colorings is again $\chi_{\text{pfo}}(\mathcal{P})$. Indeed, if G is a plane graph that realizes the latter bound in regard to χ_{pfo}, just replace each edge of G by a digon bounding a face; clearly, any facially odd coloring of the resulting multigraph is necessarily a proper facially odd coloring of the original graph G.

In the proof of our next result, Theorem 3.4, we shall use the following graph construction (see e.g. [16,20]). Given a plane multigraph G, let $v \in V_G$ be a vertex of degree $\deg_G(v) = d \geq 2$ which is not incident with a pair of parallel edges. Let the edges incident with v be enumerated in a clockwise order as $e_i = vv_i$, $i \in \mathbb{Z}_d$ (we write $\mathbb{Z}_d$ for the set $\{0, \ldots, d-1\}$ with addition modulo d); thus we assume that the vertices v_i are pairwise different. The *annihilation* of v is the construction of a plane multigraph G' from G defined as follows:

(1) add edges $e'_i = v_i v_{i+1}$, $i \in \mathbb{Z}_d$, embedded in the plane so that for each i, the edges e_i, e_{i+1}, and e'_i constitute a boundary walk of a face;
(2) remove the vertex v together with all the edges e_i.

The assumption that v is not incident with a pair of parallel edges is significant: without it, the annihilation of v may produce a loop. Intuitively, one may achieve the desired embeddings of the edges e'_i by drawing each e'_i 'close enough' to the curve consisting of the embeddings of e_i and e_{i+1}; see Fig. 4 for an example of a properly conducted annihilation of a vertex. Notice that if $d = 2$ then we introduce two parallel edges, e'_0 and e'_1, which have as endpoints the pair of vertices v_0 and v_1.

Regarding the faces of G and G', it is obvious that the following holds:

Observation 3.3. Let G' be obtained from G by the annihilation of a vertex $v \in V_G$. Then

(1) every face of G not in $F_G(v)$ is also a face of G';
(2) each face $g \in F_G(v)$ has its counterpart g' in G' such that a boundary walk of g' arises from a boundary walk of g by replacing each of its subsequences of the form $e_i v e_{i+1}$ with e'_i, and hence $V(g') = V(g) \setminus \{v\}$;
(3) there is precisely one more face in G', having as its boundary walk the sequence $v_0 e'_{d-1} v_{d-1} e'_{d-2} \cdots v_1 e'_0 v_0$.

We are ready for our main result of this section. It answers in the affirmative the concluding question of Kwon and Park [21], i.e., Question 1.3.

Theorem 3.4.

(*a*) *For every planar graph G it holds that*

$$\chi_{\text{so}}(G) \leq \chi_{\text{pfo}}(\mathcal{P}) \cdot \chi(G).$$

(*b*) *For every outerplanar graph G it holds that*

$$\chi_{\text{so}}(G) \leq \chi_{\text{pfo}}(\mathcal{P}_o) \cdot \chi(G).$$

Proof. Let $k = \chi(G)$. We consider a plane embedding of G, and denote it by G as well.

Claim 1. *There are plane multigraphs $G_1, G_2, \ldots, G_k$ whose vertex sets partition $V(G)$ into independent subsets such that for any $x \in V_G$ and any $i = 1, \ldots, k$ the following holds:*

$$\text{if } |N_G(v) \cap V(G_i)| \geq 2 \text{ then } N_G(v) \cap V_{G_i} = V(f_v) \text{ for a face } f_v \text{ of } G_i.$$

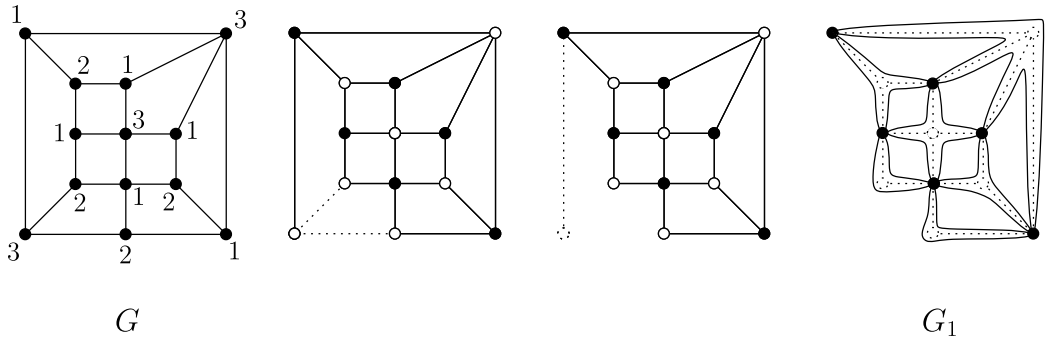

Fig. 5. From left to right, a 3-chromatic planar graph G with an optimal proper coloring, and the three-step construction of the (multi)graph G_1. The empty-circle vertices are the ones not colored by 1. In each step, the vertices/edges that are being removed appear as dotted.

Proof. Let φ be a proper coloring of G with colors $1, 2, \ldots, k$ and let $V_i = \varphi^{-1}(i)$ $(i = 1, \ldots, k)$ be the corresponding color classes. For any $i = 1, \ldots, k$ we construct the multigraph G_i from G as follows:

- first, remove every edge of G having both endpoints outside V_i;
- next, from the remaining spanning subgraph of G remove any 1^--vertex not belonging to V_i;
- finally, in any chosen order, annihilate every vertex not colored by i.

The resulting plane multigraph is G_i (cf. Fig. 5).

In view of Observation 3.3, the collection $G_1, G_2, \ldots, G_k$ meets all requirements. $\diamond$

Claim 2. *The edge set of any plane multigraph H on at least three vertices can be enlarged so that the obtained spanning supergraph $\hat{H}$ satisfies the following:*

(i) *$\hat{H}$ is a 2-connected plane multigraph; and*
(ii) *For each face f of H there exists a face $\hat{f}$ of $\hat{H}$ with $V(\hat{f}) = V(f)$.*

Proof. First we deal with the possible disconnectedness. Let the input be a plane spanning supergraph $\hat{H}$ of H so that (ii) holds. Initially, we set $\hat{H} = H$. Next, as long as the current input is disconnected, we iterate on the following. There exists a face f^* of $\hat{H}$ whose boundary $\partial(f^*)$ is disconnected. Select a pair of components, say K' and K'', of $\hat{H}$ for which it holds that $V_{K'} \cup V_{K''} \subseteq V(f^*)$. Augment $E(\hat{H})$ with a new edge e embedded within the interior of f^* so that e has one endpoint in K' and another endpoint in K''; in other words, e is a bridge linking K' and K''. Thus $\hat{H} + e$ is a plane spanning supergraph of H, and has one component fewer than $\hat{H}$. Moreover, $\hat{H} + e$ preserves the property (ii); indeed, the addition of the edge e to $E(\hat{H})$ neither destroys an existing face nor creates a new one. The graph $\hat{H} + e$ becomes our new input. Since with each iteration of the described process, the number of components of the input decreases, the end result is a connected plane multigraph, $\hat{H}$, which is a spanning supergraph of H and satisfies (ii).

Now we take care of the possible cut-vertices of the connected graph $\hat{H}$. As long as there is at least one cut-vertex, select an end-block B; say v is the cut-vertex which is incident with B. There is a face f of $\hat{H}$ such that $V(B) \cap V(f) \neq \emptyset$ and $V(B) \nsubseteq V(f)$. Thus, along any boundary walk of f, the vertex v occurs more than once. There exist vertices $u \in (V(B) \setminus \{v\}) \cap V(f)$ and $w \in V(f) \setminus V(B)$ such that a uv-edge and a vw-edge are facially adjacent in regard to f (i.e., the pair of edges are consecutive on a boundary walk of f). Augment the edge set of $\hat{H}$ with the (new) edge uw embedded within the interior of f. Thus $\hat{H} + uw$ is a plane graph and has one fewer blocks than $\hat{H}$. Additionally, $\hat{H} + uw$ preserves (ii); indeed, the face sets of $\hat{H}$ and $\hat{H} + uw$ differ only in that the face f of $\hat{H}$ is replaced by two faces f' and f'' in $\hat{H} + uw$: namely, f' is a triangular face with $V(f') = \{u, v, w\}$ and f'' is a face having the same vertex set as f (cf. Fig. 6). So the property (ii) is preserved in passing from $\hat{H}$ to $\hat{H} + uw$.

We put $\hat{H} + uw$ in place of $\hat{H}$ and iterate. Since with each iteration of the described procedure, the number of blocks of the input decreases, the end result is a 2-connected plane graph, $\hat{H}$, which is a spanning supergraph of H and satisfies (ii). $\diamond$

Let $\hat{G_1}, \hat{G_2}, \ldots, \hat{G_k}$ be the result of applying Claim 2 to the collection $G_1, G_2, \ldots, G_k$ arising from Claim 1. Take colorings φ_i of $\hat{G_i}$ $(i = 1, 2, \ldots, k)$ with pairwise disjoint color sets so that: if $\hat{G_i}$ is of order at least 3 then φ_i is facially odd, whereas if $\hat{G_i}$ is of order at most 2 then φ_i is proper. The union $\varphi = \varphi_1 \cup \varphi_2 \cup \cdots \cup \varphi_k$ is a coloring of G using at most $\chi_{\text{pfo}}(\mathcal{P}) \cdot \chi(G)$ colors. Moreover, from Claims 1 and 2 it follows that φ is a proper strong odd coloring of G. This completes the proof of (a).

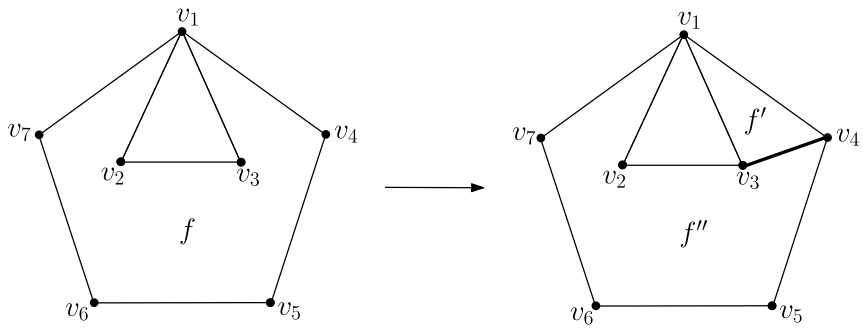

Fig. 6. If $\hat{H}$ is the graph on the left, and $B = v_1v_2v_3$ is the selected end-block (incident with the cut-vertex $v = v_1$) then such a face is f. The used boundary walk of f is $v_1v_2v_3v_1v_4v_5v_6v_7v_1$. Here, the vertices u and w are v_3 and v_4, respectively. We augment $E(\hat{H})$ with the edge v_3v_4 (depicted heavier), embedded in the interior of f. The resulting plane graph has one fewer blocks and one more face than $\hat{H}$: namely, the face f splits into a triangular face f' and a face f'' such that $V(f'') = V(f)$.

The proof of (b) goes along the same lines, completed by the observation that starting from an outerplanar G, all graphs involved in the argument remain outerplanar. □

So the question raised by the originators of strong odd colorings concerning the existence of a constant bound for the strong odd chromatic number of all planar graphs has an answer in the positive. In particular, since $\chi_{\text{pfo}}(\mathcal{P}) \leq 97$, the Four Color Theorem of Appel and Haken [2,3] and Theorem 3.4 combined yield the following corollary. The second part is derived from Wang, Finbow and Wang's theorem $\chi_{\text{pfo}}(\mathcal{P}_{\text{o}}) = 10$ from [30], using the fact that every outerplanar graph is 3-colorable.

Corollary 3.5.

(a) *For every planar graph G it holds that $\chi_{\text{so}}(G) \leq 388$.*
(b) *For every outerplanar graph G it holds that $\chi_{\text{so}}(G) \leq 30$.*

Let us define

$$\chi_{\text{so}}(\mathcal{P}) = \max_{G \hookrightarrow \mathcal{P}} \chi_{\text{so}}(G)$$

as the maximum of $\chi_{\text{so}}(G)$ over all planar graphs G. In view of Proposition 3.1 and Corollary 3.5, we have $12 \leq \chi_{\text{so}}(\mathcal{P}) \leq 388$. Similarly, let

$$\chi_{\text{so}}(\mathcal{P}_{\text{o}}) = \max_{G \text{ is outerplanar}} \chi_{\text{so}}(G)$$

be the maximum of $\chi_{\text{so}}(G)$ over all outerplanar graphs G. In view of Proposition 3.2 and Corollary 3.5, we have $7 \leq \chi_{\text{so}}(\mathcal{P}_{\text{o}}) \leq 30$. We end this section with the following.

Problem 3.6.

(1) Is $\chi_{\text{so}}(\mathcal{P}) = 12$?
(2) Is $\chi_{\text{so}}(\mathcal{P}_{\text{o}}) = 7$?

4. Strong odd colorings of graph products

In this section we study the behavior of χ_{so} with respect to all the four basic types of graph products.

Definition 4.1. Given two graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$, the vertex set in all kinds of their products discussed here is the set

$$V_G \times V_H := \{(g, h) \mid g \in V_G,\ h \in V_H\}$$

(the standard Cartesian product of the corresponding vertex sets). The edge sets are defined as follows.

(i) Cartesian product, $G \square H$:

$$E(G \square H) := \{(g, h)(g', h') \mid (g = g' \wedge hh' \in E_H) \vee (gg' \in E_G \wedge h = h'),\\ g, g' \in V_G,\ h, h' \in V_H\}.$$

(ii) Direct product, $G \times H$:

$$E(G \times H) := \{(g,h)(g',h') \mid gg' \in E_G \wedge hh' \in E_H, \\ g, g' \in V_G,\ h, h' \in V_H\}.$$

(iii) Strong product, $G \boxtimes H$:

$$E(G \boxtimes H) := \{(g,h)(g',h') \mid g' \in N[g] \wedge h' \in N[h], \\ g, g' \in V_G,\ h, h' \in V_H,\ (g,h) \neq (g',h')\}.$$

(iv) Lexicographic product, $G \circ H$:

$$E(G \circ H) := \{(g,h)(g',h') \mid gg' \in E_G \vee (g = g' \wedge hh' \in E_H), \\ g, g' \in V_G,\ h, h' \in V_H\}.$$

It turns out that χ_{so} is submultiplicative in all cases, with an interesting twist in regard to the $\circ$ operation.

Theorem 4.2. *For any two graphs G, H the following inequalities are valid.*

(i) *If $* \in \{\square, \times, \boxtimes\}$ (Cartesian, direct, or strong product), then*

$$\chi_{so}(G * H) \leq \chi_{so}(G)\, \chi_{so}(H).$$

(ii) *In case of the lexicographic product,*

$$\chi_{so}(G \circ H) \leq \chi_{so}(G)\, (\chi_{so}(H + K_1) - 1).$$

Proof. As a general notation, if F is a graph, $v \in V(F)$, and c is a color in a vertex coloring of F (any given coloring), let us write $d_{c:F}(v)$ for the number of neighbors of v in color c. If $v = (g,h)$ is a product vertex, we simplify notation and write $d_{c:F}(g,h)$ instead of $d_{c:F}((g,h))$. Similar convention will be applied to color assignments as well, writing $\varphi(g,h)$ for $\varphi((g,h))$.

For any of the four products we let φ_G be any strong odd coloring of G with $\chi_{so}(G)$ colors. Similarly, for the three product types of (i), let φ_H be any strong odd coloring of H with $\chi_{so}(H)$ colors. For (ii) we consider a strong odd coloring of $H + K_1$ with $\chi_{so}(H + K_1)$ colors, and let φ_H be its restriction to V_H. This latter φ_H uses $\chi_{so}(H + K_1) - 1$ colors because the color of K_1 does not appear in V_H.

In either case, we compose a coloring of the product graph by the rule

$$\varphi(g,h) := (\varphi_G(g), \varphi_H(h))$$

for all $g \in V_G$, $h \in V_H$. We claim that every vertex has an odd (or zero) number of neighbors in each color.

In $G \square H$ the neighbors of any vertex (g,h) are of the form (g,h') or (g',h), implying that the first or second coordinate of the color at each neighbor coincides with the color at the vertex in question. Moreover, if $(\varphi_G(g), \varphi_H(h)) = (\varphi_G(g'), \varphi_H(h'))$, then (g,h) and (g',h') are not adjacent. Indeed, if $g = g'$, then $\varphi_H(h) = \varphi_H(h')$ implies $hh' \notin E_H$; and the reasoning is analogous if $h = h'$. So, for any composite color $c = (c_G, c_H)$, the neighbors of any vertex (g,h) assigned with c belong entirely to a copy of G or to a copy of H in $G \square H$. It follows that the number of those neighbors is equal either to $d_{c_G:G}(g)$ or to $d_{c_H:H}(h)$, hence odd or zero.

In $G \times H$ if (g,h) and (g',h') are adjacent, we necessarily have $\varphi_G(g) \neq \varphi_G(g')$ and $\varphi_H(h) \neq \varphi_H(h')$. Then, for any composite color $c = (c_G, c_H)$, the equality $d_{c:G\times H}(g,h) = d_{c_G:G}(g)\, d_{c_H:H}(h)$ is valid. Consequently $d_{c:G\times H}(g,h)$ is either zero or the product of two odd numbers, hence odd.

The open neighborhood $N_{G\boxtimes H}(g,h)$ of a vertex (g,h) in the strong product is the disjoint union $N_{G\square H}(g,h) \cup N_{G\times H}(g,h)$. As observed above, the composite color of (g,h) in $G \square H$ shares its first or second coordinate with the colors of its neighbors, which is not the case with neighbors in $G \times H$. Thus, $\varphi(N_{G\square H}(g,h)) \cap \varphi(N_{G\times H}(g,h)) = \varnothing$, and consequently the strong odd coloring property of φ is inherited from those of φ_G and φ_H. This completes the proof of (i).

For (ii) we first note that

$$V(G \circ H) = \bigcup_{g \in V_G} (\{g\} \times H);$$

moreover, if $gg' \in E_G$ then the copies $\{g\} \times H$ and $\{g'\} \times H$ of H are completely adjacent, and if $gg' \notin E_G$ then they are completely nonadjacent. Furthermore, in case of adjacency we have $\varphi(\{g\} \times H) \cap \varphi(\{g'\} \times H) = \varnothing$.

Assume that (g,h) has the composite color $c = (c_G, c_H)$ in $G \circ H$, and consider its neighbors (g',h') of an arbitrary specified color $c' = (c'_G, c'_H)$. We need to distinguish between two cases.

If $c'_G = c_G$, then we must have $g' = g$. Indeed, the color class c_G is an independent set in G, and for $g' \neq g$ the two copies of H are nonadjacent, cannot contain any color-c' neighbor of (g, h). Thus, the number of neighbors of (g, h) in color c' is equal to the number of color-c'_H neighbors of vertex h in graph H, which is odd, due to φ_H.

On the other hand, if $c'_G \neq c_G$, then we claim

$$d_{c':G\circ H}(g, h) = d_{c'_G:G}(g) \cdot |\{h' \in V_H : \varphi_H(h') = c'_H\}| .$$

Indeed, the first term on the right-hand side is the number of adjacent copies of H where color c'_G occurs as first coordinate of the composite color; this number is odd, due to φ_G. The second term is of course just the number of vertices in each copy of H, whose second color-coordinate is c'_H. It is exactly the number of color-c'_H neighbors of K_1 in $H + K_1$; and this is also odd, because φ_H is derived from a strong odd coloring. Thus, $d_{c':G\circ H}(g, h)$ is odd, proving (ii). □

Remark 4.3. The outerplanar example in Section 3 shows that the insertion of a universal vertex may drastically increase the strong odd chromatic number; i.e., $\chi_{so}(H + K_1) - 1$ may be substantially larger than $\chi_{so}(H)$, and the upper bound in (ii) may not be replaced with the formula of (i). Indeed, $\chi_{so}(P_6) = 3$, while $\chi_{so}(P_6 + K_1) = 7$.

Tightness of Theorem 4.2 is clear for strong and lexicographic products, since $K_p \boxtimes K_q \cong K_p \circ K_q \cong K_{pq}$ holds for all natural numbers p and q. The situation with the other two types of products is more interesting. The next theorem disregards the trivial case of $q = 1$ since $K_p \square K_1 \cong K_p$ and $K_p \boxtimes K_1$ is the edgeless graph $\overline{K}_p \cong pK_1$. The formula given in part (i) below is explicitly stated for the case of $p = q$ in the Introduction of [21], and in fact the principle mentioned there applies also for $p \neq q$.

Theorem 4.4. *Let $p, q \geq 2$ be integers.*

(i) *We have $\chi_{so}(K_p \square K_q) = pq$ for all p, q.*
(ii) *If both p and q are odd (and at least 3), then also $\chi_{so}(K_p \times K_q) = pq$.*
(iii) *If precisely one of p and q is even, say p is even and q is odd, then $\chi_{so}(K_p \times K_q) = q$.*
(iv) *If both p and q are even, then $\chi_{so}(K_p \times K_q) = \min(p, q)$.*

Proof. Let $V(K_p) = V_G = \{g_1, \ldots, g_p\}$ and $V(K_q) = V_H = \{h_1, \ldots, h_q\}$. We refer to the sets $\{g_i\} \times V_H$ $(1 \leq i \leq p)$ as *columns*, and to $V_G \times \{h_j\}$ $(1 \leq i \leq q)$ as *rows*.

In $K_p \square K_q$ no row and no column can contain any color more than once, in any proper vertex coloring. Moreover, if a color occurs in both column i and row j, then vertex (g_i, h_j) has precisely two neighbors of that color in $K_p \square K_q$. Consequently, no repeated color can occur in any strong odd coloring. This proves (i).

Turning to $K_p \times K_q$, a vertex coloring is proper if and only if each of its color classes is contained entirely in a row or in a column. We claim the following.

(v) *In any strong odd coloring of $K_p \times K_q$, if p is odd, then no row can contain a color more than once; and similarly, if q is odd, then no column can contain a color more than once.*

Since $K_p \times K_q \cong K_q \times K_p$, it suffices to prove (v) for rows. Suppose for argument's sake that for some $r \geq 2$ the vertices $(g_1, h_1), (g_2, h_1), \ldots, (g_r, h_1)$ have the same color c, but none of the vertices (g_ℓ, h_1) have color c for $r < \ell \leq p$. If $r = p$, then vertex (g_1, h_2) has exactly $p - 1$ neighbors of color c, which is even. Otherwise, if $r < p$, then (g_r, h_2) has $r - 1 \geq 1$ neighbors of color c, and (g_{r+1}, h_2) has r neighbors of color c; and one of them is even. Hence, in either case color c occurs an even number of times in the neighborhood of at least one vertex, proving (v).

Observe that (v) implies (ii), because if both p and q are odd, then no color can be repeated at all.

If q is odd, then (v) implies that exactly q colors are used in each column. But for p even, q colors are also enough, because we can then make each row monochromatic, assigning color j to all vertices in $V_G \times \{h_j\}$ $(j = 1, 2, \ldots, q)$. In that case each vertex outside of row j has exactly $p - 1$ neighbors of color j. This proves (iii).

Finally, if both p and q are even, then making each row or each column monochromatic, we obtain a strong odd coloring with $\min(p, q)$ colors. This is optimal because $\chi_{so}(K_p \times K_q) \geq \chi(K_p \times K_q) \geq pq/\alpha(K_p \times K_q) = pq/\max(p, q) = \min(p, q)$. This proves (iv). □

Remark 4.5. The parameter χ_{so} is not multiplicative with respect to the Cartesian product operation. As an example, consider $F = G \square H$, where $G = g_1 g_2 g_3 g_4 g_5 \cong C_5$ and $H = h_1 h_2 h_3 h_4 h_5 \cong C_5$. The following table exhibits a strong odd coloring of F with five colors.

	h_1	h_2	h_3	h_4	h_5
g_1	1	2	3	4	5
g_2	4	5	1	2	3
g_3	2	3	4	5	1
g_4	5	1	2	3	4
g_5	3	4	5	1	2

Each cell (g_i, h_j) indicates the color of the corresponding vertex. Here the closed neighborhood of each vertex contains each color exactly once. □

4.1. Nordhaus–Gaddum-type consequences

From the above we can derive that there do not exist any nontrivial lower or upper bounds for the strong odd chromatic number of a graph and its complement as a function of the number of vertices. We express this fact in the following result.

Theorem 4.6. *Let $n = (2k+1)^2$ be an odd square.*

(i) There exists a graph H_1 of order n such that both $\chi_{so}(H_1) = \sqrt{n}$ and $\chi_{so}(\overline{H_1}) = \sqrt{n}$.
(ii) There exists a graph H_2 of order n such that both $\chi_{so}(H_2) = n$ and $\chi_{so}(\overline{H_2}) = n$.
(iii) For every graph G on n vertices,

$$2\sqrt{n} \le \chi_{so}(G) + \chi_{so}(\overline{G}) \le 2n\,,$$
$$n \le \chi_{so}(G) \cdot \chi_{so}(\overline{G}) \le n^2,$$

and all these inequalities are tight whenever n is an odd square.

Proof. We can take $H_1 = (2k+1)K_{2k+1}$, whose complement is the complete $(2k+1)$-partite graph $K_{2k+1,\dots,2k+1}$. Each component K_{2k+1} of H_1 has a strong odd coloring from the color set $\{1, 2, \dots, 2k+1\}$. Also, assigning color i to all vertices of the ith component, a strong odd coloring of $\overline{H_1}$ with $2k+1 = \sqrt{n}$ colors is obtained. This proves (i).

Further, we can take $H_2 = K_{2k+1} \square K_{2k+1}$, whose complement is $\overline{H_2} = K_{2k+1} \times K_{2k+1}$. Then, by parts (i) and (ii) of Theorem 4.4, we have $\chi_{so}(H_2) = \chi_{so}(\overline{H_2}) = n$. This proves (ii).

The lower bounds in (iii) follow from the additive and multiplicative versions of the Nordhaus–Gaddum theorem on the chromatic number, since $\chi_{so}(G) \ge \chi(G)$ holds for all graphs G by definition. The upper bounds in (iii) are clear as the obvious inequality $\chi_{so}(G) \le |V(G)|$ is valid for all G. Tightness follows from the constructions given for (i) and (ii). □

Remark 4.7. Sometimes $\chi_{so}(G)$ behaves nearly as $\chi(G^2)$, as shown by the following two examples.

- The product graph $G = K_p \square K_p$ with $n = p^2$ vertices satisfies $\chi_{so}(G) + \chi_{so}(\overline{G}) = 2n$. We also note that it is regular of degree $2p-2$, hence $\chi_{so}(G) = p^2 \approx (\Delta(G))^2/4$.
- Strongly regular graphs $G \in \mathrm{Srg}(n, k, \lambda, \mu)$ with appropriate choices of the parameters have the property that both G and $\overline{G}$ have diameter 2, hence if $|G| = n$ then $\chi(G^2) + \chi((\overline{G})^2) = 2n$. Note, however, that G^2 and $(\overline{G})^2$ are not complementary to each other.

We close this paper with the following question.

Problem 4.8. Does $\chi_{so}(G) \le (1 + o(1))\Delta^2/4$ hold for every graph G of maximum degree Δ, as Δ gets large?

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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